

ch 8 - Depreciation & Taxes

X _____ X

Depreciation is ↓se in monetary value over period of time.

Causes

- Tear & Wear
- Effusion of Time
- Depletion
- Fall in Market Value

• Method of Depreciation Calculation:

A. St Line Method

- Linear / Uniform / Constnt Rate

- Annual Depreciation (D_n) = $\frac{I - S}{N}$
per year
- Total Depreciation value (D) = $I - S$
- Rate of Depreciation (R) = $\frac{1}{N} \times 100\%$

B. Declining Balance / Matheson / Diminishing

- Constant % $\downarrow$ use

• Annual Depreciation $(D_n) = R I (1-R)^{n-1}$

C. SOYD (Sum of Year Digit)

$$SOYD = \sum_{i=1}^n i = \frac{N(N+1)}{2}$$

N = Useful Life

n = given year

$$\bullet \text{ Annual SOYD Depreciation } (D_n) = \frac{I-S}{SOYD} (N-n+1)$$

D. MACRS (Modified Accelerated Cost Recovery System)

- Half year convention

- Salvage value $(S) = 0$

Given, MACRS Table

• After Tax Cash Flow (ATCF)

$$ATCF = \text{net income} + \text{Depreciation} + \text{Amortization}$$

Given, R_k = Revenue during k

E_k = Expense during k

d_k = Depreciation amount during k

t = Income tax

Calculate,

$$\text{Taxable amount } (T_k) = R_k - E_k - d_k$$

$$\text{Tax amount } (T_{ak}) = t T_k$$

$$BTCF_k = R_k - E_k$$

$$ATCF_k = BTCF_k - T_{ak}$$

Tabular,

Year	BTCF	depreciation	Taxable amount (T_k)	Tax amount (T_{ak})	ATCF
k	$R_k - E_k$	d_k	$BTCF - d_k$	$t T_k$	$BTCF - t T_k$